

MATRICES

1. Definition

A rectangular arrangement of numbers in rows and columns, is called a matrix. Such a rectangular arrangement of numbers is enclosed by small () or big [] brackets. Generally a matrix is represented by a capital letter A, B, C..... etc. and its elements are represented by small letters a, b, c, x, y etc.

Following are some examples of a matrix :

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{bmatrix} 1 & 5 & 3 \\ 4 & 0 & 2 \end{bmatrix}$$

$$C = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad D = [1, 5, 6], \quad E = [5]$$

2. Order of Matrix

A matrix which has m rows and n columns is called a matrix of order $m \times n$, and it is represented by

$$A_{m \times n} \text{ or } A = [a_{ij}]_{m \times n}$$

It is obvious to note that a matrix of order $m \times n$ contains mn elements. Every row of such a matrix contains n elements and every column contains m elements.

3. Representation of a Matrix

A matrix of order $m \times n$ is generally expressed as

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1j} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2j} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{i1} & a_{i2} & \dots & a_{ij} & \dots & a_{in} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mj} & \dots & a_{mn} \end{bmatrix}$$

$$\text{or } A = [a_{ij}]_{m \times n} \quad i = 1, 2, \dots, m$$

$$\text{or } A = [a_{ij}] \quad j = 1, 2, \dots, n$$

From this representation, it is clear that (i, j)th element of a matrix A is written as a_{ij}

$$\text{It may also be written as } (A)_{ij}. \text{ e.g. } \begin{bmatrix} 1 & 5 & 7 \\ -2 & 6 & 3 \\ 2 & 1 & 9 \end{bmatrix}$$

$$a_{11} = 1, a_{12} = 5, a_{13} = 7; a_{21} = -2, a_{22} = 6, a_{23} = 3; a_{31} = 2, a_{32} = 1, a_{33} = 9$$

4. Types of Matrices

4.1 Row matrix

If in a matrix, there is only one row, then it is called a Row Matrix.

Thus $A = [a_{ij}]_{m \times n}$ is a row matrix if $m = 1$

eg. $[1, 3, 5]$ is a row matrix of order 1×3

4.2 Column Matrix

If in a matrix, there is only one column, then it is called a column matrix.

Thus $A = [a_{ij}]_{m \times n}$ is a column matrix if $n = 1$.

eg. $\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$ is column matrix of order 3×1 .

4.3 Square matrix

If number of rows and number of column in a matrix are equal, then it is called a square matrix.

Thus $A = [a_{ij}]_{m \times n}$ is a square matrix if $m = n$.

eg. $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ is a square matrix of order 3×3 .

Note : (a) If $m \neq n$ then matrix is called a rectangular matrix.

(b) The elements of a square matrix A for which $i = j$ i.e., $a_{11}, a_{22}, a_{33}, \dots, a_{nn}$ are called diagonal elements and the line joining these elements is called the principal diagonal or of leading diagonal of matrix A .

(c) **Trace of a matrix :** The sum of diagonal elements of a square matrix. A is called the trace of matrix A which is denoted by trace A .

$$\text{trace } A = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \dots + a_{nn}$$

4.4 Singleton matrix

If in a matrix there is only one element then it is called singleton matrix.

Thus

$A = [a_{ij}]_{m \times n}$ is a singleton matrix if $m = n = 1$.

eg. $[4], [2], [b], [-5]$ are singleton matrices.

4.5 Null or zero matrix

If in a matrix all the elements are zero then it is called a zero matrix and it is generally denoted by O .

Thus $A = [a_{ij}]_{m \times n}$ is a zero matrix if $a_{ij} = 0$ for all i and j .

eg. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is a zero matrix of order 3×3 .

4.6 Diagonal matrix

If all elements except the principal diagonal in a square matrix are zero, it is called a diagonal matrix.

Thus a square matrix

$A = [a_{ij}]$ is a diagonal matrix if $a_{ij} = 0$, when $i \neq j$.

eg. $\begin{bmatrix} 5 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 7 \end{bmatrix}$ is a diagonal matrix of order 3×3 , which also can be denoted by $\text{diag } [5, 6, 7]$

Note : (a) No element of principal diagonal in diagonal matrix is zero.

(b) Number of zero in a diagonal matrix is given by $n^2 - n$ where n is the order of the matrix.

4.7 Scalar Matrix

If all the elements of the diagonal of a diagonal matrix are equal, it is called a scalar matrix.

Thus a square matrix $A [a_{ij}]$ is a scalar matrix is

$$a_{ij} = \begin{cases} 0 & i \neq j \\ k & i = j \end{cases} \text{ where } k \text{ is a constant.}$$

$$\text{eg. } \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \text{ is a scalar matrix.}$$

4.8 Unit matrix

If all elements of principal diagonal in a **diagonal matrix** are 1, then it is called unit matrix. A unit matrix of order n is denoted by I_n .

Thus a square matrix

$A = [a_{ij}]$ is a unit matrix if

$$a_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

$$\text{eg. } I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Note : Every unit matrix is a scalar matrix.

4.9 Triangular matrix

A square matrix $[a_{ij}]$ is said to be triangular if each element above or below the principal diagonal is zero it is of two types -

(a) Upper triangular matrix : A square matrix $[a_{ij}]$ is called the upper triangular matrix, if $a_{ij} = 0$ when $i > j$.

$$\text{eg. } \begin{bmatrix} 4 & 2 & 5 \\ 0 & 6 & 7 \\ 0 & 0 & 4 \end{bmatrix} \text{ is a upper triangular matrix of order } 3 \times 3$$

(b) Lower triangular matrix : A square matrix $[a_{ij}]$ is called the lower triangular matrix, if

$a_{ij} = 0$ when $i < j$

$$\text{eg. } \begin{bmatrix} 7 & 0 & 0 \\ 4 & 9 & 0 \\ 2 & 5 & 2 \end{bmatrix} \text{ is a lower triangular matrix of order } 3 \times 3.$$

Note : Minimum number of zero in a triangular matrix is given by $\frac{n(n-1)}{2}$ where n is order of matrix.

4.10 Equal matrix

Two matrix A and B are said to be equal matrix if they are of same order and their corresponding elements are equal

$$\text{eg. if } A = \begin{bmatrix} 2 & 3 & 4 \\ 6 & 5 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix} \text{ are equal matrix then}$$

$$a_1 = 2, a_2 = 3, a_3 = 4, b_1 = 6, b_2 = 5, b_3 = 1$$

4.11 Singular matrix

Matrix A is said to be singular matrix if its determinant $|A| = 0$, otherwise non-singular matrix i.e.,

If $\det |A| = 0 \Rightarrow$ singular

and $\det |A| \neq 0 \Rightarrow$ non-singular

5. Addition and subtraction of Matrices

If $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ are two matrices of the same order then their sum $A + B$ is a matrix whose each element is the sum of corresponding element.

i.e., $A + B = [a_{ij} + b_{ij}]_{m \times n}$

eg. If $A = \begin{bmatrix} 7 & 3 \\ 4 & 2 \\ 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 2 \\ 5 & 4 \\ 3 & 7 \end{bmatrix}$ then $A + B = \begin{bmatrix} 7+6 & 3+2 \\ 4+5 & 2+4 \\ 2+3 & 1+7 \end{bmatrix} = \begin{bmatrix} 13 & 5 \\ 9 & 6 \\ 5 & 8 \end{bmatrix}$

then $A - B = \begin{bmatrix} 7-6 & 3-2 \\ 4-5 & 2-4 \\ 2-3 & 1-7 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \\ -1 & -6 \end{bmatrix}$

Similarly their subtraction $A - B$ is defined as

$$A - B = [a_{ij} - b_{ij}]_{m \times n}$$

i.e., in above example

$$A - B = \begin{bmatrix} 5-1 & 2-5 \\ 1-2 & 3-2 \\ 4-3 & 1-3 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -1 & 1 \\ 1 & -2 \end{bmatrix}$$

Note : Matrix addition and subtraction can be possible only when matrices are of same order.

5.1 Properties of matrices addition

If A, B and C are matrices of same order, then-

(i) $A + B = B + A$ (Commutative Law)

(ii) $(A + B) + C = A + (B + C)$ (Associative law)

(iii) $A + O = O + A = A$, where O is zero matrix which is additive identity of the matrix.

(iv) $A + (-A) = 0 = (-A) + A$ where $(-A)$ is obtained by changing the sign of every element of A which is additive inverse of the matrix

$$(v) \left. \begin{array}{l} A + B = A + C \\ B + A = C + A \end{array} \right\} \Rightarrow B = C \text{ (Cancellation law)}$$

(vi) $\text{trace}(A + B) = \text{trace}(A) \pm \text{trace}(B)$

6. Scalar Multiplication of Matrices

Let $A = [a_{ij}]_{m \times n}$ be a matrix and k be a number then the matrix which is obtained by multiplying every element of A by k is called scalar multiplication of A by k and it denoted by kA .

Thus

$$A = [a_{ij}]_{m \times n} \Rightarrow kA = [ka_{ij}]_{m \times n}$$

e.g. if $A = \begin{bmatrix} 4 & 2 \\ 3 & 5 \\ 6 & 7 \end{bmatrix}$ then $5A = \begin{bmatrix} 20 & 10 \\ 15 & 25 \\ 30 & 35 \end{bmatrix}$

6.1 Properties of scalar multiplication

If A, B are matrices of the same order and m, n are any numbers, then the following results can be easily established.

$$(i) m(A + B) = mA + mB$$

$$(ii) (m + n)A = mA + nA$$

$$(iii) m(nA) = (mn)A = n(mA)$$

Ex.1 If $\begin{bmatrix} 1 & 0 \\ 3 & -4 \end{bmatrix} + \begin{bmatrix} a & 1 \\ -1 & b \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 2 & -2 \end{bmatrix}$, then value of a, b are-

[1] 1, -2

[2] -1, 2

[3] -1, -2

[4] 1, 2

Sol. Here

$$\begin{bmatrix} 1+a & 0+1 \\ 3-1 & -4+b \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 2 & -2 \end{bmatrix}$$

$$\Rightarrow 1 + a = 2 \text{ and } -4 + b = -2 \Rightarrow a = 1, b = 2$$

Ex.2 If $X = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ and $3X - \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$ then a is equal to-

[1] 1

[2] 2

[3] 0

[4] -2

Sol. $3X = \begin{bmatrix} 3 & 3a \\ 0 & 3 \end{bmatrix}$

$$\Rightarrow \text{L.H.S.} = \begin{bmatrix} 3-2 & 3a-3 \\ 0-0 & 3-2 \end{bmatrix} = \begin{bmatrix} 1 & 3a-3 \\ 0 & 1 \end{bmatrix}$$

No by equality of two matrices, we have $3a - 3 = 3 \Rightarrow a = 2$.

Ex.3 If X and Y two matrices are such that $X - Y = \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix}$ and $X + Y = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ then Y matrices is

[1] $\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$

[2] $\begin{bmatrix} -1 & -2 \\ 3 & 4 \end{bmatrix}$

[3] $\begin{bmatrix} -1 & -2 \\ 2 & 2 \end{bmatrix}$

[4] none of these

Sol. Given that $X - Y = \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix}$ (i)

and $X + Y = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ (ii)

Subtracting (2) from (1)

$$-2Y = \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix} - \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$$

$$(-2)Y = \begin{bmatrix} 3-1 & 2-(-2) \\ -1-3 & 0-4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ -4 & -4 \end{bmatrix}$$

$$\Rightarrow Y = -\frac{1}{2} \begin{bmatrix} 2 & 4 \\ -4 & -4 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 2 & 2 \end{bmatrix}$$

Ex.4 A matrix $A = [a_{ij}]$ of order 2×3 whose elements are such that $a_{ij} = i + j$ is-

- [1] $\begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$ [2] $\begin{bmatrix} 2 & 3 & 4 \\ 5 & 4 & 3 \end{bmatrix}$ [3] $\begin{bmatrix} 2 & 3 & 4 \\ 5 & 5 & 4 \end{bmatrix}$ [4] none of these

Sol. a_{ij} is the element of i^{th} row and j^{th} column of matrix A

$$\therefore \begin{aligned} a_{11} &= 1 + 1 = 2, a_{12} = 1 + 2 = 3, a_{13} = 1 + 3 = 4 \\ a_{21} &= 2 + 1 = 3, a_{22} = 2 + 2 = 4, a_{23} = 2 + 3 = 5 \end{aligned}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

Ex.5 If $A = \begin{bmatrix} 1 & 3 \\ 3 & 2 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix}$ and $A + B - D = 0$ (zero matrix), then D matrix will be-

- [1] $\begin{bmatrix} 0 & 2 \\ 3 & 7 \\ 6 & 5 \end{bmatrix}$ [2] $\begin{bmatrix} 0 & 2 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$ [3] $\begin{bmatrix} 0 & 1 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$ [4] $\begin{bmatrix} 0 & -2 \\ -3 & -7 \\ -5 & -6 \end{bmatrix}$

Sol. Let $D = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$

$$\therefore A + B - D = \begin{bmatrix} 1 & 3 \\ 3 & 2 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1-1-a & 3-2-b \\ 3+0-c & 2+5-d \\ 2+3-e & 5+1-f \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{aligned} \Rightarrow -a &= 0 \Rightarrow a = 0, & 1-b &= 0 \Rightarrow b = 1, \\ 3-c &= 0 \Rightarrow c = 3, & 7-d &= 0 \Rightarrow d = 7, \\ 5-e &= 0 \Rightarrow e = 5, & 6-f &= 0 \Rightarrow f = 6 \end{aligned}$$

$$\therefore D = \begin{bmatrix} 0 & 1 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$$

Ex.6 If $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & k & 5 \\ 4 & 2 & 1 \end{bmatrix}$ is a singular matrix then k is equal to-

- [1] -1 [2] 8 [3] 4 [4] -8

Sol. A is singular $\Rightarrow |A| = 0$

$$\Rightarrow \begin{vmatrix} 1 & -3 & 2 \\ 2 & k & 5 \\ 4 & 2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow 1(k - 10) + 3(2 - 20) + 2(4 - 4k) = 0$$

$$\Rightarrow 7k + 56 = 0 \Rightarrow k = -8$$

7. Multiplication of matrices

If A and B be any two matrices, then their product AB will be defined only when number of column in A is equal to the number of rows in B. If $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{n \times p}$ then their product $AB = C = [c_{ij}]$, will be matrix of order $m \times p$, where

$$(AB)_{ij} = C_{ij}$$

$$= \sum_{r=1}^n a_{ir} b_{rj}$$

eg. If $A = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 2 & 2 \\ 1 & 3 \end{bmatrix}$

then $AB = \begin{bmatrix} 1 \cdot 1 + 4 \cdot 2 + 2 \cdot 1 & 1 \cdot 2 + 4 \cdot 2 + 2 \cdot 3 \\ 2 \cdot 1 + 3 \cdot 2 + 1 \cdot 1 & 2 \cdot 2 + 3 \cdot 2 + 1 \cdot 3 \end{bmatrix}$

$$AB = \begin{bmatrix} 11 & 16 \\ 9 & 13 \end{bmatrix}$$

7.1 Properties of matrix multiplication

If A, B and C are three matrices such that their product is defined, then

- (i) $AB \neq BA$ (Generally not commutative)
- (ii) $(AB)C = A(BC)$ (Associative Law)
- (iii) $IA = A = AI$ I is identity matrix for matrix multiplication
- (iv) $A(B + C) = AB + AC$ (Distributive law)
- (v) If $AB = AC \nRightarrow B = C$ (cancellation Law is not applicable)
- (vi) If $AB = 0$ It does not mean that $A = 0$ or $B = 0$, again product of two non-zero matrix may be zero matrix.
- (vii) $\text{trance}(AB) = \text{trance}(BA)$

- Note :**
- (i) The multiplication of two diagonal matrices is again a diagonal matrix.
 - (ii) The multiplication of two triangular matrices is again a triangular matrix.
 - (iii) The multiplication of two scalar matrices is also a scalar matrix.
 - (iv) If A and B are two matrices of the same order, then
 - (a) $(A + B)^2 = A^2 + B^2 + AB + BA$
 - (b) $(A - B)^2 = A^2 + B^2 - AB - BA$
 - (c) $(A - B)(A + B) = A^2 - B^2 + AB - BA$
 - (d) $(A + B)(A - B) = A^2 - B^2 - AB + BA$
 - (e) $A(-B) = (-A)B = -(AB)$

7.2 Positive Integral powers of a matrix

The positive integral powers of a matrix A are defined only when A is a square matrix.

Also then $A^2 = A.A$ $A^3 = A.A.A = A^2A$

Also for any positive integers m, n

- (i) $A^m A^n = A^{m+n}$ (ii) $(A^m)^n = A^{mn} = (A^n)^m$
- (iii) $I^n = I, I^m = I$ (iv) $A^0 = I_n$ where A is a square matrices of order n.

Ex.7 If matrix $P = \begin{bmatrix} 0 & -\tan \phi/2 \\ \tan \phi/2 & 0 \end{bmatrix}$ then $(I - P) \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$ is equal to-

- [1] p [2] $P + I$ [3] I [4] none of these

Sol.

$$I - P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & -\tan \phi/2 \\ \tan(\phi/2) & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & +\tan(\phi/2) \\ -\tan(\phi/2) & 1 \end{bmatrix}$$

$$\therefore (I - P) \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \tan(\phi/2) \\ -\tan(\phi/2) & 1 \end{bmatrix} \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$$

$$= \begin{bmatrix} \cos \phi + \tan(\phi/2)\sin \phi & -\sin \phi + \tan(\phi/2)\cos \phi \\ -\tan(\phi/2)\cos \phi + \sin \phi & \tan(\phi/2)\sin \phi + \cos \phi \end{bmatrix}$$

$$= \begin{bmatrix} 1 - 2\sin^2(\phi/2) + 2\sin^2(\phi/2) & -2\sin(\phi/2)\cos(\phi/2) + \tan(\phi/2)(2\cos^2(\phi/2) - 1) \\ -\tan(\phi/2)(2\cos^2(\phi/2) - 1) + 2\sin(\phi/2)\cos(\phi/2) & \tan(\phi/2)(2\sin(\phi/2)\cos(\phi/2)) + (1 - 2\sin^2(\phi/2)) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -\tan(\phi/2) \\ \tan(\phi/2) & 1 \end{bmatrix} = I + P$$

Ex.8 If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and $A^2 - 4A - nI = 0$, then n is equal to-

- [1] 3 [2] -3 [3] 1/3 [4] -1/3

Sol. $A^2 = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$, $4A = \begin{bmatrix} 8 & -4 \\ -4 & 8 \end{bmatrix}$, $nI = \begin{bmatrix} n & 0 \\ 0 & n \end{bmatrix}$

$$\Rightarrow A^2 - 4A - nI$$

$$= \begin{bmatrix} 5 - 8 - n & -4 + 4 - 0 \\ -4 + 4 - 0 & 5 - 8 - n \end{bmatrix}$$

$$= \begin{bmatrix} -3 - n & 0 \\ 0 & -3 - n \end{bmatrix}$$

$$\therefore A^2 - 4A - nI = 0$$

$$\Rightarrow \begin{bmatrix} -3 - n & 0 \\ 0 & -3 - n \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow -3 - n = 0$$

$$\Rightarrow n = -3$$

Ex.9 If $[1 \times 2] \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ 1 \\ -1 \end{bmatrix} = 0$, then the value of x is-

[1] -1

[2] 0

[3] 1

[4] 2

Sol. The LHS of the equation

$$= [2 \ 4x + 9 \ 2x + 5] \begin{bmatrix} x \\ 1 \\ -1 \end{bmatrix}$$

$$= [2x + 4x + 9 - 2x - 5] = 4x + 4$$

$$\text{Thus } 4x + 4 = 0 \Rightarrow x = -1$$

Ex.10 If $E(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$, then value of $E(\alpha) \cdot E(\beta)$ is-

[1] $E(0^\circ)$

[2] $E(90^\circ)$

[3] $E(\alpha + \beta)$

[4] $E(\alpha - \beta)$

Sol. $E(\alpha) \cdot E(\beta)$

$$= \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \beta & \sin \beta \\ -\sin \beta & \cos \beta \end{bmatrix}$$

$$= \begin{bmatrix} \cos \alpha \cos \beta - \sin \alpha \sin \beta & \cos \alpha \sin \beta + \sin \alpha \cos \beta \\ -\sin \alpha \cos \beta - \cos \alpha \sin \beta & -\sin \alpha \sin \beta + \cos \alpha \cos \beta \end{bmatrix}$$

$$= \begin{bmatrix} \cos(\alpha + \beta) & \sin(\alpha + \beta) \\ -\sin(\alpha + \beta) & \cos(\alpha + \beta) \end{bmatrix} = E(\alpha + \beta)$$

Ex.11 If $A = \begin{bmatrix} -1 & 2 \\ 3 & -4 \end{bmatrix}$ then element a_{21} of A^2 is-

[1] 22

[2] -15

[3] -10

[4] 7

Sol. The element a_{21} is product of second row of A to the first column of A

$$\therefore a_{21} = [3 \ -4] \begin{bmatrix} -1 \\ 3 \end{bmatrix} = -3 - 12 = -15$$

Ex.12 If $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$ then

[1] $x = 2, y = 1$

[2] $x = 1, y = 2$

[3] $x = 3, y = 2$

[4] $x = 2, y = 3$

Sol. The given matrix equation can be written as

$$\begin{bmatrix} x + 2y \\ 2x + y \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

$$\Rightarrow x + 2y = 5 \text{ and } 2x + y = 4$$

$$\Rightarrow x = 1, y = 2$$

8. Transpose of Matrix

If we interchange the rows and column of a matrix A, then the matrix so obtained is called the transpose of A and it is denoted by

$$A^T \text{ or } A^t \text{ or } A'$$

From this definition it is obvious to note that

(i) Order of A is $m \times n \Rightarrow$ order of A^T is $n \times m$

(ii) $(A^T)_{ij} = (A)_{ji}, \forall i, j$

8.1 Properties of Transpose

If A, B are matrices of suitable order then

(i) $(A^T)^T = A$

(ii) $(A + B)^T = A^T + B^T$

(iii) $(A - B)^T = A^T - B^T$

(iv) $(kA)^T = kA^T$

(v) $(AB)^T = B^T A^T$

(vi) $(A_1 A_2 \dots A_n)^T = A_n^T \dots A_2^T A_1^T$

(vii) $(A^n)^T = (A^T)^n, n \in \mathbb{N}$

Ex.13 If $A = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 1 \\ 7 & 2 \end{bmatrix}$ then $B^T A^T$ is equal to-

[1] $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

[2] $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

[3] $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

[4] $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Sol. $B^T A^T = \begin{bmatrix} 4 & 7 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -7 \\ -1 & 4 \end{bmatrix}$

$$= \begin{bmatrix} 8-7 & -28+28 \\ 2-2 & -7+8 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Ex.14 If $A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 1 & 6 \end{bmatrix}$ then $(AB)^T$ equals-

[1] $\begin{bmatrix} 5 & 16 \\ 9 & 16 \end{bmatrix}$

[2] $\begin{bmatrix} 5 & 9 \\ 16 & 12 \end{bmatrix}$

[3] $\begin{bmatrix} 5 & 9 \\ 4 & 3 \end{bmatrix}$

[4] none of these

Sol. $AB = \begin{bmatrix} 3+2 & 4+12 \\ 9+0 & 12+0 \end{bmatrix}$

$$= \begin{bmatrix} 5 & 16 \\ 9 & 12 \end{bmatrix}$$

$$\therefore (AB)^T = \begin{bmatrix} 5 & 9 \\ 16 & 12 \end{bmatrix}$$

9. Symmetric and skew-symmetric Matrix

(a) Symmetric matrix : A square matrix $A = [a_{ij}]$ is called symmetric matrix if $a_{ij} = a_{ji}$ for all i, j or $A^T = A$.

$$\text{eg. } \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$$

Note : (i) Every unit matrix and square zero matrix are symmetric matrices.

(ii) Maximum number of different element in a symmetric matrix is $\frac{n(n+1)}{2}$

(b) Skew-symmetric matrix : A square matrix $A = [a_{ij}]$ is called skew-symmetric matrix if

$$a_{ij} = -a_{ji} \text{ for all } i, j$$

$$\text{or } A^T = -A \quad \text{eg. } \begin{bmatrix} 0 & h & g \\ -h & 0 & f \\ -g & -f & 0 \end{bmatrix}$$

Note : (i) All principal diagonal elements of a skew-symmetric matrix are always zero because for any diagonal element -

$$a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0$$

(ii) Trace of a skew symmetric matrix is always 0

9.1 Properties of symmetric and skew-symmetric matrices

(i) If A is a square matrix, then $A + A^T$, AA^T , $A^T A$ are symmetric matrices while $A - A^T$ is skew-symmetric matrices.

(ii) If A, B are two symmetric matrices, then-

(a) $A \pm B$, $AB + BA$ are also symmetric matrices.

(b) $AB - BA$ is a skew-symmetric matrix.

(c) AB is a symmetric matrix when $AB = BA$

(iii) If A, B are two skew-symmetric matrices, then-

(a) $A \pm B$, $AB - BA$ are skew-symmetric matrices.

(b) $AB + BA$ is a symmetric matrix.

(iv) If A is a skew-symmetric matrix and C is a column matrix, then $C^T A C$ is a zero matrix.

(v) Every square matrix A can uniquely be expressed as sum of a symmetric and skew symmetric matrix i.e.,

$$A = \left[\frac{1}{2}(A + A^T) \right] + \left[\frac{1}{2}(A - A^T) \right]$$

Ex.15 If $A = \begin{bmatrix} -1 & 7 \\ 2 & 3 \end{bmatrix}$, then skew-symmetric part of A is-

$$[1] \begin{bmatrix} -1 & 9/2 \\ -9/2 & 3 \end{bmatrix}$$

$$[2] \begin{bmatrix} 0 & -5/2 \\ 5/2 & 0 \end{bmatrix}$$

$$[3] \begin{bmatrix} -1 & -9/2 \\ 9/2 & 3 \end{bmatrix}$$

$$[4] \begin{bmatrix} 0 & 5/2 \\ -5/2 & 0 \end{bmatrix}$$

Sol. Let $A = B + C$, where $B = \frac{1}{2}(A + A^T)$ and $C = \frac{1}{2}(A - A^T)$ are respectively symmetric and skew symmetric part of A .

$$\text{Now } C = \frac{1}{2} \left\{ \begin{bmatrix} -1 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 7 & 3 \end{bmatrix} \right\} = \frac{1}{2} \begin{bmatrix} 0 & 5 \\ -5 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 5/2 \\ -5/2 & 0 \end{bmatrix}$$

10. Determinant of a Matrix

If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ be a square matrix, then its determinant, denoted by $|A|$ or $\det. (A)$ is

defined as

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

10.1 Properties of the determinant of a matrix

- (i) $|A|$ exist $\Leftrightarrow A$ is a square matrix
- (ii) $|AB| = |A| |B|$
- (iii) $|A^T| = |A|$
- (iv) $|kA| = k^n |A|$, if A is a square matrix of order n .
- (v) If A and B are square matrices of same order then $|AB| = |BA|$
- (vi) If A is skew symmetric matrix of odd order then $|A| = 0$
- (vii) If $A = \text{diag} (a_1 a_2, \dots, a_n)$ then $|A| = a_1 a_2 \dots a_n$
- (viii) $|A|^n = |A^n|$, $n \in \mathbb{N}$

11. Adjoint of a Matrix

If every element of a square matrix A be replaced by its cofactor in $|A|$, then the transpose of the matrix so obtained is called the adjoint of A and it is denoted by $\text{adj } A$

Thus if $A = [a_{ij}]$ be a square matrix and F_{ij} be the cofactor of a_{ij} in $|A|$, then

$$\text{adj } A = [F_{ij}]^T$$

$$\Rightarrow (\text{adj } A)_{ij} = F_{ij}$$

Hence if $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$, then

$$\text{adj } A = \begin{bmatrix} F_{11} & F_{12} & \dots & F_{1n} \\ F_{21} & F_{22} & \dots & F_{2n} \\ \dots & \dots & \dots & \dots \\ F_{n1} & F_{n2} & \dots & F_{nn} \end{bmatrix}^T = \begin{bmatrix} F_{11} & F_{21} & \dots & F_{n1} \\ F_{12} & F_{22} & \dots & F_{n2} \\ \dots & \dots & \dots & \dots \\ F_{1n} & F_{2n} & \dots & F_{nn} \end{bmatrix}$$

11.1 Properties of Adjoint Matrix

If A, B are square matrices of order n and I_n is corresponding unit matrix, then

$$(i) \quad A (\text{adj } A) = |A| I_n = (\text{adj } A) A$$

(Thus $A (\text{adj } A)$ is always a scalar matrix)

$$(ii) \quad |\text{adj } A| = |A|^{n-1}$$

$$(iii) \quad \text{adj} (\text{adj } A) = |A|^{n-2} A$$

$$(iv) \quad |\text{adj} (\text{adj } A)| = |A|^{(n-1)^2}$$

$$(v) \quad \text{adj} (A^T) = (\text{adj } A)^T$$

$$(vi) \quad \text{adj} (AB) = (\text{adj } B) (\text{adj } A)$$

$$(vii) \quad \text{adj} (A^m) = (\text{adj } A)^m, m \in \mathbb{N}$$

$$(viii) \quad \text{adj} (kA) = k^{n-1} (\text{adj } A), k \in \mathbb{R}$$

$$(ix) \quad \text{adj} (I_n) = I_n$$

$$(x) \quad \text{adj } 0 = 0$$

$$(xi) \quad A \text{ is symmetric} \Rightarrow \text{adj } A \text{ is also symmetric.}$$

$$(xii) \quad A \text{ is diagonal} \Rightarrow \text{adj } A \text{ is also diagonal.}$$

$$(xiii) \quad A \text{ is triangular} \Rightarrow \text{adj } A \text{ is also triangular.}$$

$$(xiv) \quad A \text{ is singular} \Rightarrow |\text{adj } A| = 0$$

Ex.16 If $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$, then $|\text{adj} (\text{adj } A)|$ is equal -

[1] 8

[2] 16

[3] 2

[4] 0

Sol. $|A| = \begin{vmatrix} 1 & 0 & 3 \\ 2 & 1 & 1 \\ 0 & 0 & 2 \end{vmatrix} = 2$

$$\therefore |\text{adj} (\text{adj } A)| = |A|^{(n-1)^2} = |A|^{2^2} \quad [\because \text{Here } n = 3]$$

$$= 2^4 = 16$$

Ex.17 If $A = \begin{bmatrix} 1 & 3 & 5 \\ 3 & 5 & 1 \\ 5 & 1 & 3 \end{bmatrix}$, then $\text{adj. } A$ is equal to-

[1] $\begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$ [2] $\begin{bmatrix} -14 & 4 & 22 \\ 4 & 22 & -14 \\ 22 & -14 & 4 \end{bmatrix}$ [3] $\begin{bmatrix} 14 & 4 & -22 \\ 4 & -22 & -14 \\ -22 & -14 & -4 \end{bmatrix}$ [4] none of these

Sol. $\text{adj. } A = \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}^T = \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$

Ex.18 If $A = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}$, then $\text{adj}(\text{adj } A)$ is equal to-

[1] $8 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

[2] $16 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

[3] $64 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

[4] none of these

Sol. $|A| = \begin{vmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{vmatrix} = (2)(2)(2) = 8$

now $\text{adj}(\text{adj } A) = |A|^{3-2} A$

$= 8 \begin{vmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{vmatrix} = 16 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

12. Inverse Matrix

If A and B are two matrices such that

$$AB = I = BA$$

then B is called the inverse of A and it is denoted by A^{-1} . Thus

$$A^{-1} = B \Leftrightarrow AB = I = BA$$

Further we may note from above property (i) of adjoint matrix that if $|A| \neq 0$, then

$$A \frac{\text{adj}(A)}{|A|} = I = \frac{(\text{adj } A)}{|A|} A$$

$$\Rightarrow A^{-1} = \frac{1}{|A|} \text{adj } A$$

Thus A^{-1} exists $\Leftrightarrow |A| \neq 0$.

Note :

(i) Matrix A is called invertible if A^{-1} exists.

(ii) Inverse of a matrix is unique.

12.1 Properties of Inverse Matrix

(i) $(A^{-1})^{-1} = A$

(ii) $(A^T)^{-1} = (A^{-1})^T$

(iii) $(AB)^{-1} = B^{-1}A^{-1}$

(iv) $(A^n)^{-1} = (A^{-1})^n, n \in \mathbb{N}$

(v) $\text{adj}(A^{-1}) = (\text{adj } A)^{-1}$

(vi) $|A^{-1}| = \frac{1}{|A|} = |A|^{-1}$

(vii) $A = \text{diag}(a_1, a_2, \dots, a_n) \Rightarrow A^{-1} = \text{diag}(a_1^{-1}, a_2^{-1}, \dots, a_n^{-1})$

- (viii) A is symmetric $\Rightarrow A^{-1}$ is also symmetric.
 (ix) A is diagonal $|A| \neq 0 \Rightarrow A^{-1}$ is also diagonal.
 (x) A is scalar matrix $\Rightarrow A^{-1}$ is also scalar matrix.
 (xi) A is triangular $|A| \neq 0 \Rightarrow A^{-1}$ is also triangular.

Ex.19 Inverse matrix of $\begin{bmatrix} 2 & -3 \\ -4 & 2 \end{bmatrix}$ is-

[1] $-\frac{1}{8} \begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$ [2] $-\frac{1}{8} \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$ [3] $\frac{1}{8} \begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$ [4] $\begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$

Sol. Let the given matrix is A , then $|A| = -8$

and $\text{adj } A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}^T = \begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$

$\therefore A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{8} \begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$

Ex.20 If $A = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $M = AB$, then M^{-1} is equal to-

[1] $\begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix}$ [2] $\begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 1/6 \end{bmatrix}$ [3] $\begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 1/6 \end{bmatrix}$ [4] $\begin{bmatrix} 1/3 & -1/3 \\ -1/3 & 1/6 \end{bmatrix}$

Sol. $M = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -2 & 2 \end{bmatrix}$

$|M| = 6$, $\text{adj } M = \begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix}$

$\therefore M^{-1} = \frac{1}{6} \begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 1/6 \end{bmatrix}$

13. Some important cases of Matrices

13.1 Orthogonal Matrix

A square matrix A is called orthogonal if

$AA^T = I = A^T A$; i.e., if $A^{-1} = A^T$

eg. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is a orthogonal matrix because here $A^{-1} = A^T$

13.2 Idempotent matrix

A square matrix A is called an idempotent matrix if

$$A^2 = A$$

eg. $\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$ is a idempotent matrix because here $A^2 = A$

13.3 Involutory Matrix

A square matrix A is called an involutory matrix if

$$A^2 = I \quad \text{or} \quad A^{-1} = A$$

eg. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is a involutory matrix.

13.4 Nilpotent matrix

A square matrix A is called a nilpotent matrix if there exist a $p \in \mathbb{N}$ such that

$$A^p = 0$$

eg. $A = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$ is a nilpotent matrix.

13.5 Hermition matrix

A square matrix A is skew-Hermition matrix if

$$A^{\theta} = A \quad ; \quad \text{i.e.,} \quad a_{ij} = -\bar{a}_{ji} \quad \forall i, j$$

13.6 Skew hermitian matrix

A square matrix A is skew-hermitian is

$$A = -A^{\theta}$$

i.e., $a_{ij} = -\bar{a}_{ji} \quad \forall i, j$

13.7 Period of a matrix

If for any matrix A

$$A^{k+1} = A$$

then k is called period of matrix (where k is a least positive integer)

eg. If $A^3 = A$, $A^5 = A$, $A^7 = A$,.....then it is a period matrix and $A^{2+1} = A$ so its period is = 2

13.8 Differentiation of matrix

If $A = \begin{bmatrix} f(x) & g(x) \\ h(x) & \ell(x) \end{bmatrix}$

then $\frac{dA}{dx} = \begin{bmatrix} f'(x) & g'(x) \\ h'(x) & \ell'(x) \end{bmatrix}$ is a differentiation of matrix A

eg. if $A = \begin{bmatrix} x^2 & \sin x \\ 2x & 2 \end{bmatrix}$ then $\frac{dA}{dx} = \begin{bmatrix} 2x & \cos x \\ 2 & 0 \end{bmatrix}$

13.9 Submatrix

Let A be $m \times n$ matrix, then a matrix obtained by leaving some rows or columns or both of a is called a sub matrix of A

eg. if $A' = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 2 & 2 \\ 2 & 5 & 3 \end{bmatrix}$ and $\begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix}$ are sub matrices of matrix $A = \begin{bmatrix} 2 & 1 & 0 & -1 \\ 3 & 2 & 2 & 4 \\ 2 & 5 & 3 & 1 \end{bmatrix}$

13.10 Rank of a matrix

A number r is said to be the rank of a $m \times n$ matrix A if

- (a) every square sub matrix of order $(r + 1)$ or more is singular and
- (b) there exists at least one square submatrix of order r which is non-singular.

Thus, the rank of matrix is the order of the highest order non-singular sub matrix.

eg. The rank of matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 3 & 4 & 5 \end{bmatrix}$ is

We have $|A| = 0$ therefore $r(A)$ is less than 3, we observe that $\begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix}$ is a non-singular square sub matrix of order 2 hence $r(A) = 2$.

Note :

- (i) The rank of the null matrix is not defined and the rank of every non null matrix is greater than or equal to one.
- (ii) The rank of matrix is same as the rank of its transpose i.e., $r(A) = r(A^T)$
- (iii) Elementary transformation of not alter the rank of matrix.